$m\frac{l^2}{T^2}$ G O G

B

O

Oxy MN x $t = 0$

D_1 D_2 C_1 $t = 3T/4$

D_1 C_1 $t = 10T/4$ D_2

G O

Q (3 cm,3 cm) O

Q Oxy MN

a、b、c、d



m a、b l k_1

c、d l k_2

b、c

c、d a、b v_0

1. $k_1 = k_2$

2. $k_1 = 4k_2$

6. (25 points)

Blocks a、b、c、d, of mass m, lie on a smooth level table. A spring of natural length l joins a、b; its stiffness is k_1. A spring of natural length l, stiffness k_2, joins c、d. All four centres lie on a line. b、c collisions are perfectly elastic and very brief. Both springs initially have natural length; c、d rest, while a、b move right at the same velocity v_0. Discuss quantitatively:

(1) For $k_1=k_2$, how do the four blocks move relative to the table?

(2) For $k_1=4k_2$, how do the four blocks move relative to the table?